RSA Public-Key Encryption

26. Write a C program for RSA public-key encryption scheme, each user has a public key, e, and aprivate key, d. Suppose Bob leaks his private key. Rather than generating

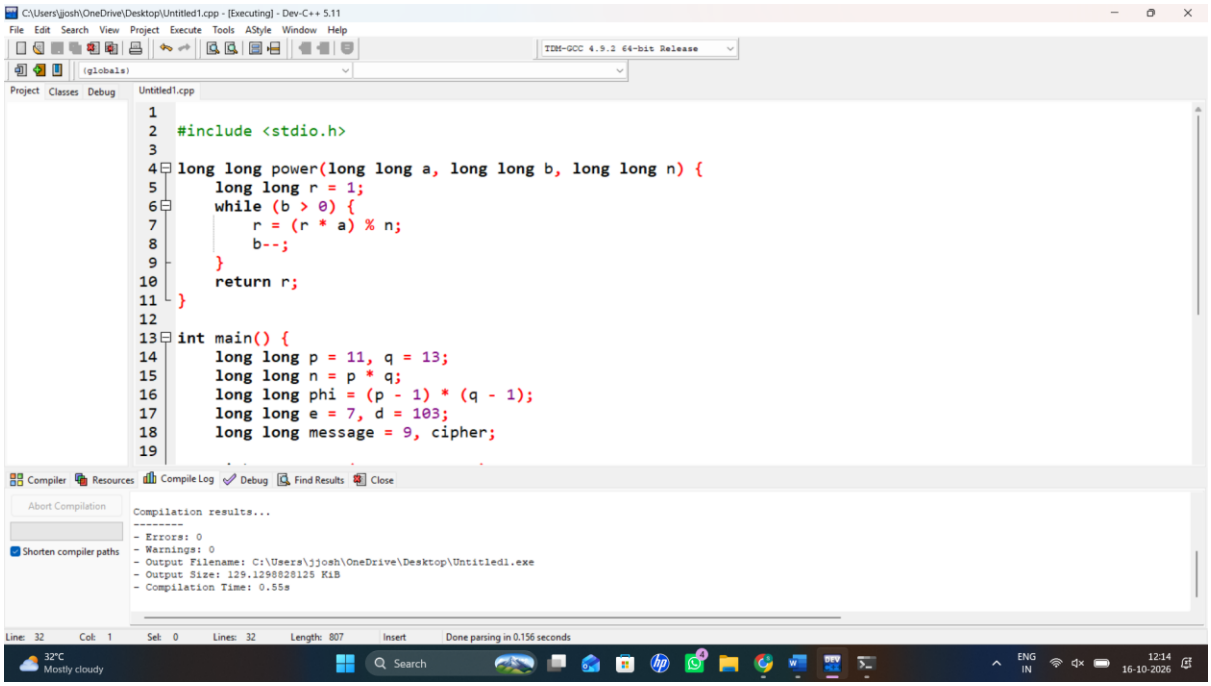
Aim:

To implement the RSA public-key encryption scheme in C and demonstrate why generating new public and private keys without changing a compromised modulus is unsafe.

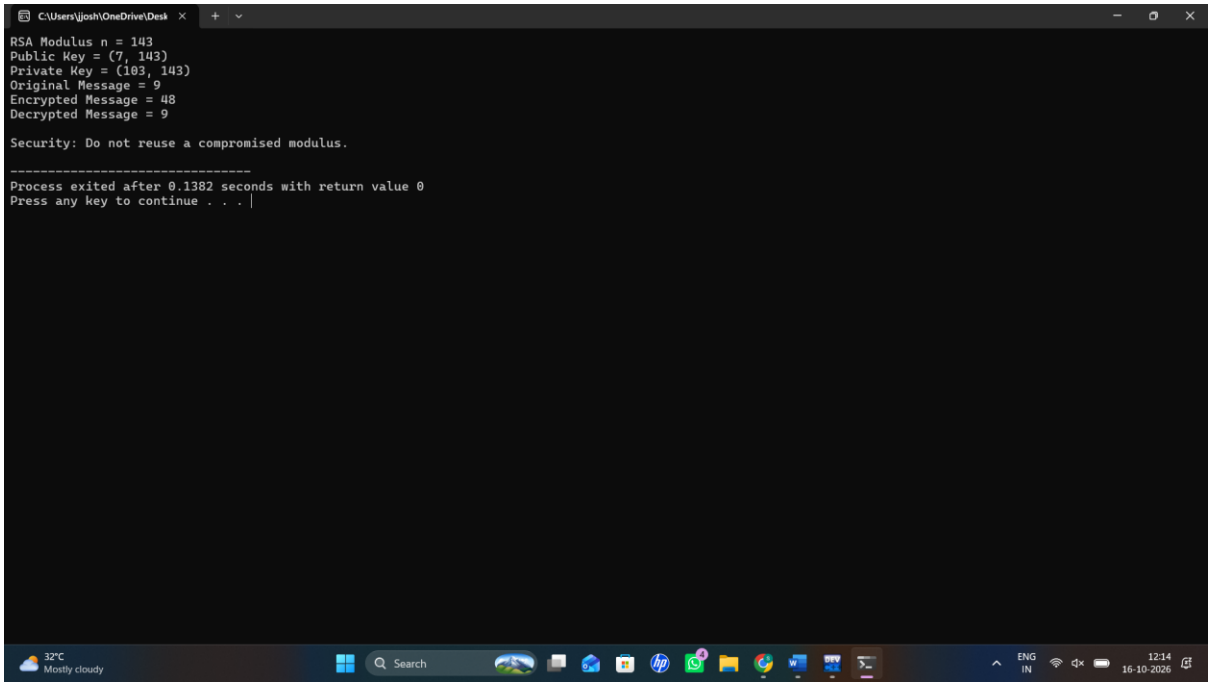
Algorithm:

1. Select two prime numbers \(p\) and \(q\).
2. Calculate \(n=pq\) and \(\phi(n)=(p-1)(q-1)\).
3. Select public exponent \(e\) and calculate private exponent \(d\).
4. Encrypt the message using \(C=M^e\bmod n\).
5. If the private key is leaked, generate another key pair using the same modulus.
6. Observe that the compromised modulus may allow an attacker to recover the new private key.

Program:



Output:



Result:

The RSA encryption and decryption operations were implemented successfully. Generating new exponents while reusing a compromised modulus is unsafe because an attacker who knows its factorization can calculate the new private key.